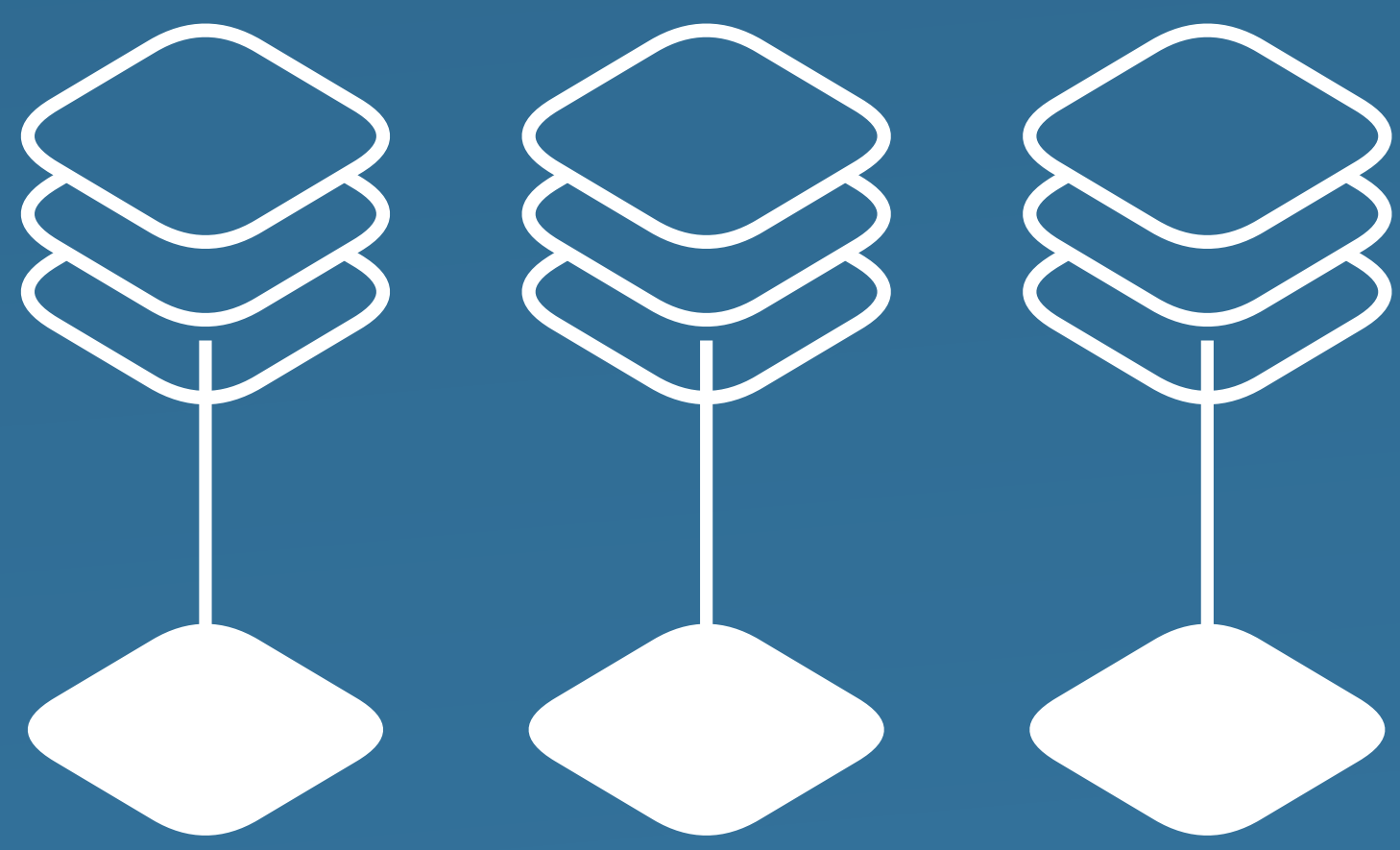




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, each NYUv2 task peaks at a different depth

≈22×
fewer params than concatenating layers (ViT-L, dense tasks)

Let's talk!
Open to research collaborations
vision foundation models · representation learning · probing & interpretability · spatial understanding



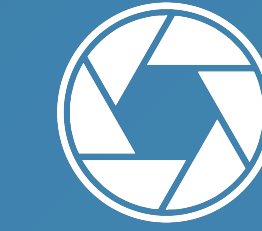
turhancankargin.me

AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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Copernicus Science Centre
Warsaw · October 8–11, 2026

Tasks live at different depths

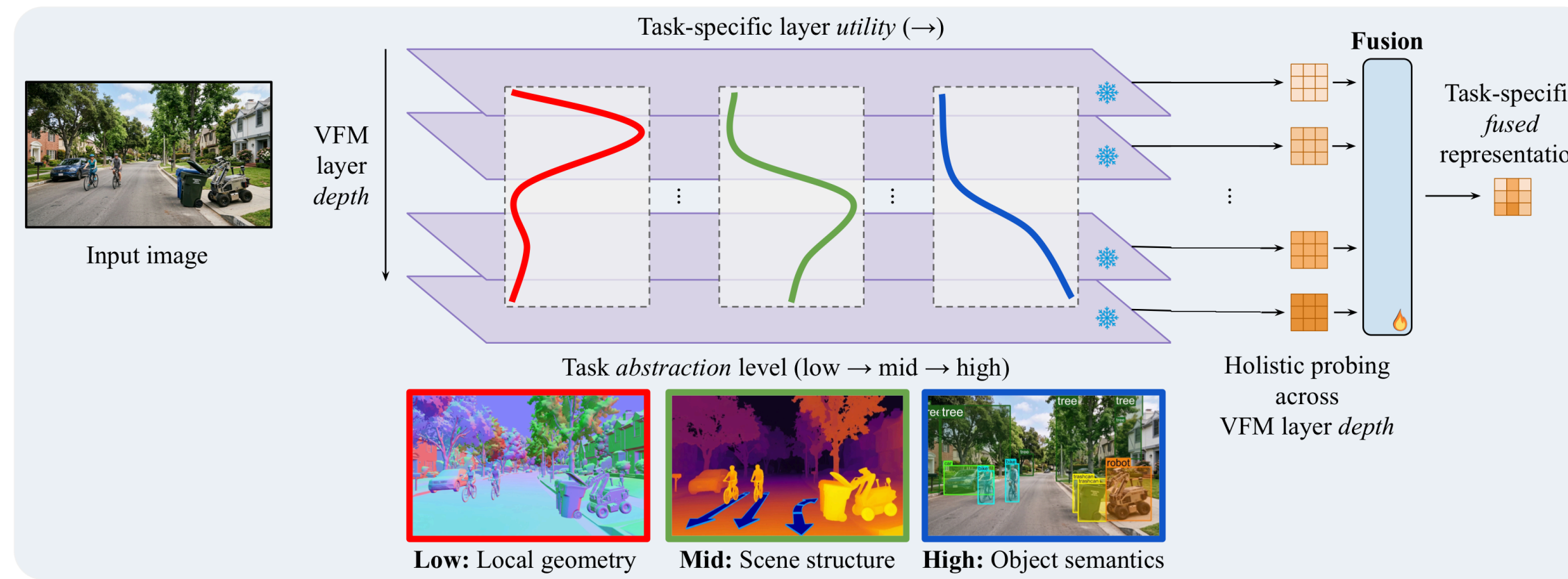


Fig. 1: Geometry peaks mid-depth, semantics at the end; AMDF fuses all layers per patch.

#1: last layer ≠ always best

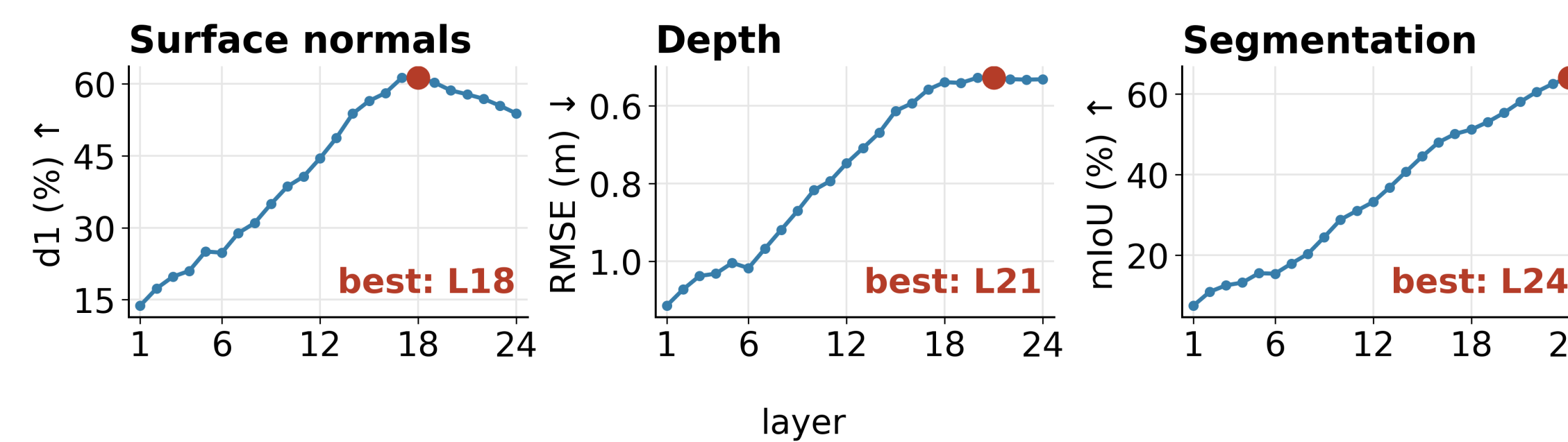


Fig. 2: Best single layer differs per task: ≈L18 (normals), ≈L21 (depth), L24 (segmentation); one seed per layer (DINOv3-L, NYUv2).

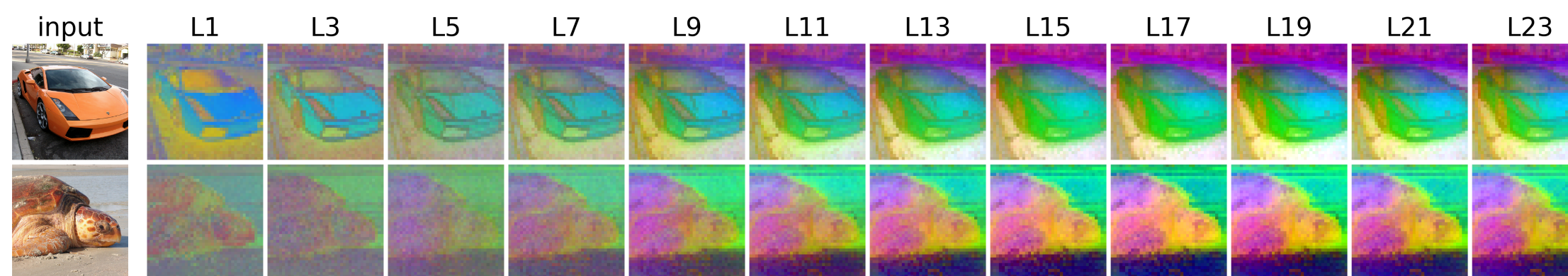


Fig. 3: MAE-L, ImageNet: **fine detail** early, **object regions** and parts (turtle shell vs head) deep.

#2: all layers = costly

✗ Concat all layers: 26.2M params, the same fixed weights for every patch.

✓ AMDF: per-patch layer choice, ≈1.2M params.

Take-home

- Useful features are **spread across depth**.
- AMDF: **2nd-best mean rank** with **≈22×** fewer params than Concat (ViT-L).
- AMDF beats every **fixed layer mix** by mean rank.

7 frozen VFMs

5 benchmarks

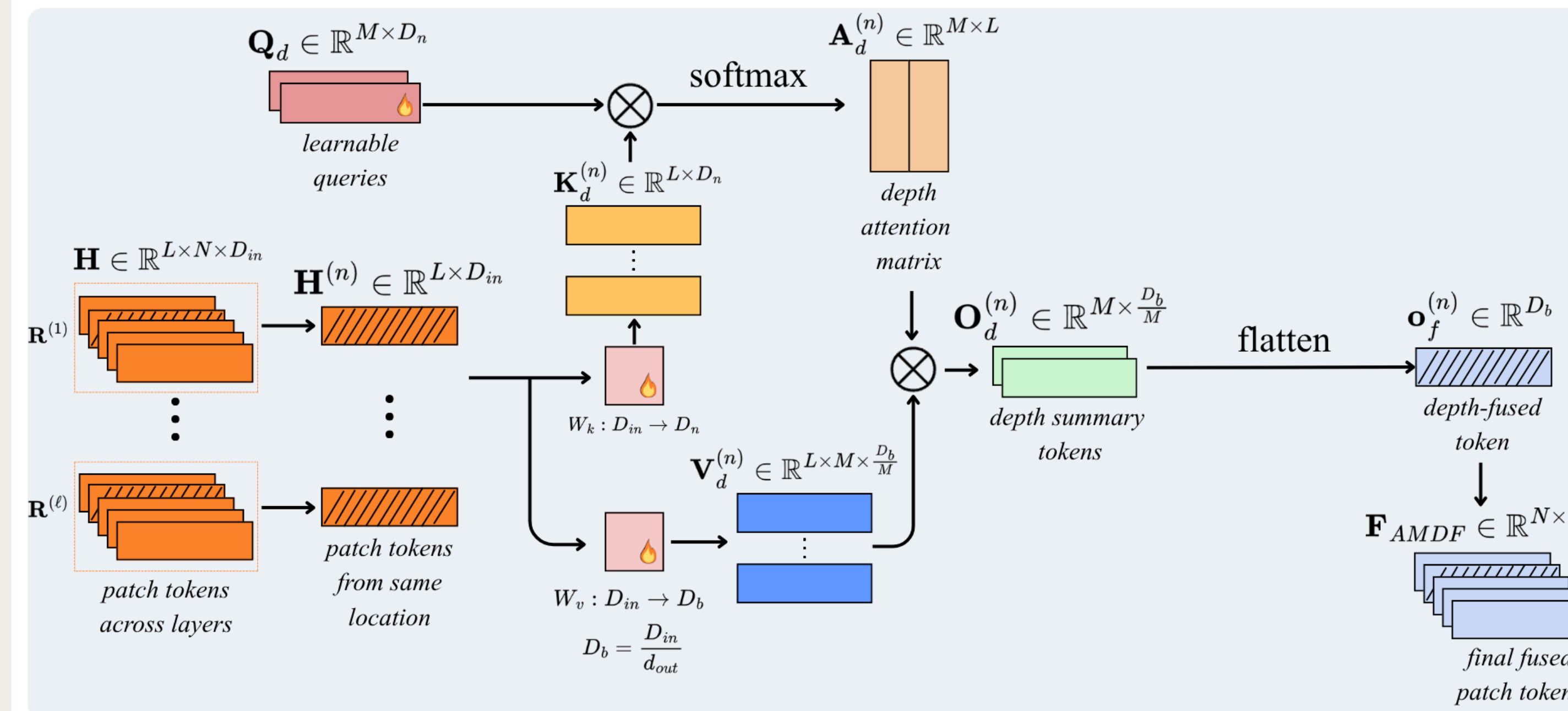
5 baselines

3 seeds

DINOv3-S/B/L · MAE-L · SigLIP2-L · V-JEPA2.1-L · PESpatial-L

Solution: AMDF — depth-aware fusion

- Per patch, learnable queries score all layers
- Softmax over depth → a task-specific layer mix
- Frozen backbone, light task heads



Attention vs. layer utility (earlier variant)

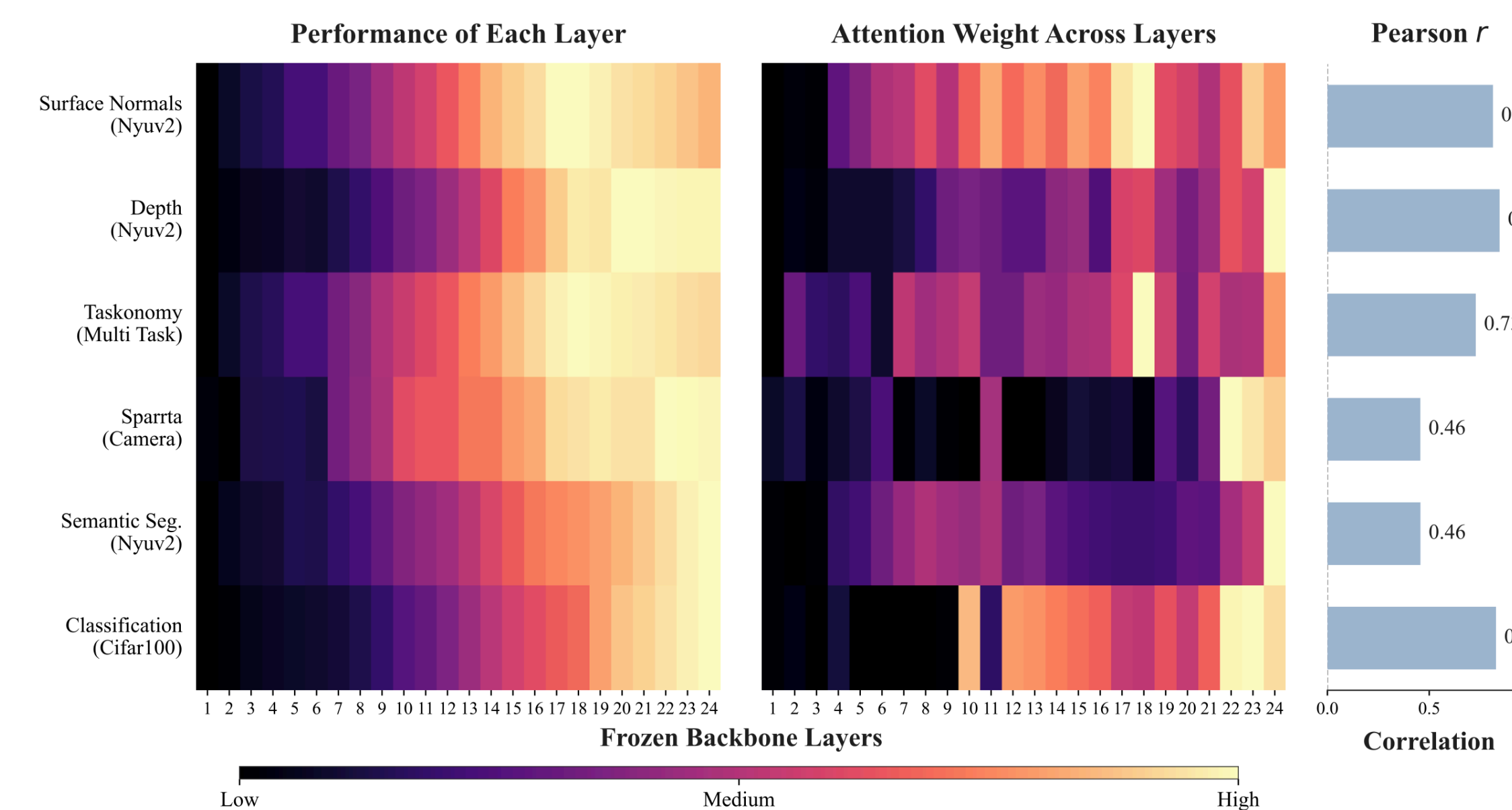


Fig. 4: Earlier AMDF variant (DINOv3-L): attention correlates with per-layer probe utility, $r = 0.46$ – 0.85 across 6 tasks.

What AMDF predicts

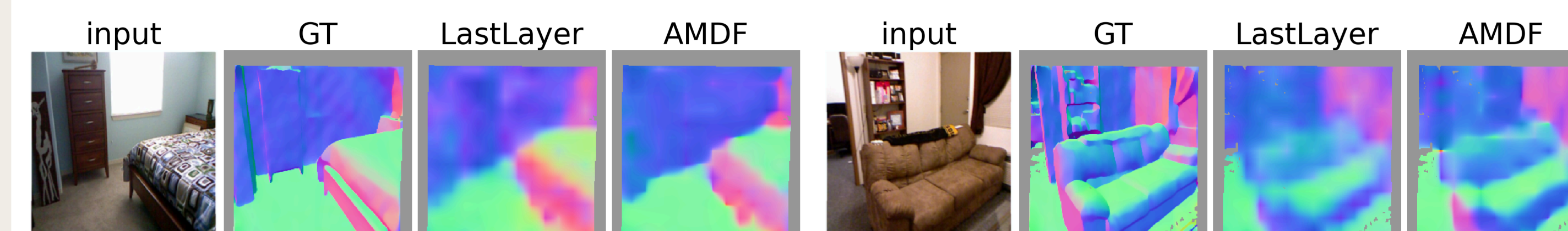


Fig. 5: NYUv2 normals, DINOv3-L (seed-8 retrain), selected examples (gain at the 83rd / 52nd percentile): AMDF recovers **sharper geometry** than the last-layer probe.

Per-layer queries: better mean on all 5 tasks

Normals	Taskonomy	Segm.	Depth	Pose
+1.62 d1	-0.73 L1	+0.16 mIoU	+0.51 d1	-2.37° rot.
7/7 backbones	7/7 backbones	6/7 backbones	5/7 backbones	5/7 backbones

Paired gains vs. shared queries, mean over 7 backbones; Taskonomy L1 ×100; cost: +2% params.

Results

≈22× fewer params than Concat (ViT-L)
2nd mean rank (1st on depth)
5×7 benchmarks × backbones

Mean rank per benchmark

	Normals	Depth	Segm.	Taskonomy	Pose
AMDF (ours)	2.00	1.43 ★	1.86	2.00	1.93
ConcatLayers	1.00 ★	1.71	1.57 ★	1.00 ★	1.21 ★
WeightedLayers	4.14	4.00	4.14	3.29	3.89
SpatialWeights	3.14	4.14	4.43	4.00	4.00
LastLayer	5.43	4.14	3.86	5.29	4.96
AvgLayer	5.29	5.57	5.14	5.43	5.00

Fig. 6: Mean rank per benchmark (★ best). Concat 1st on 4/5, AMDF 1st on depth.

Accuracy vs. parameters

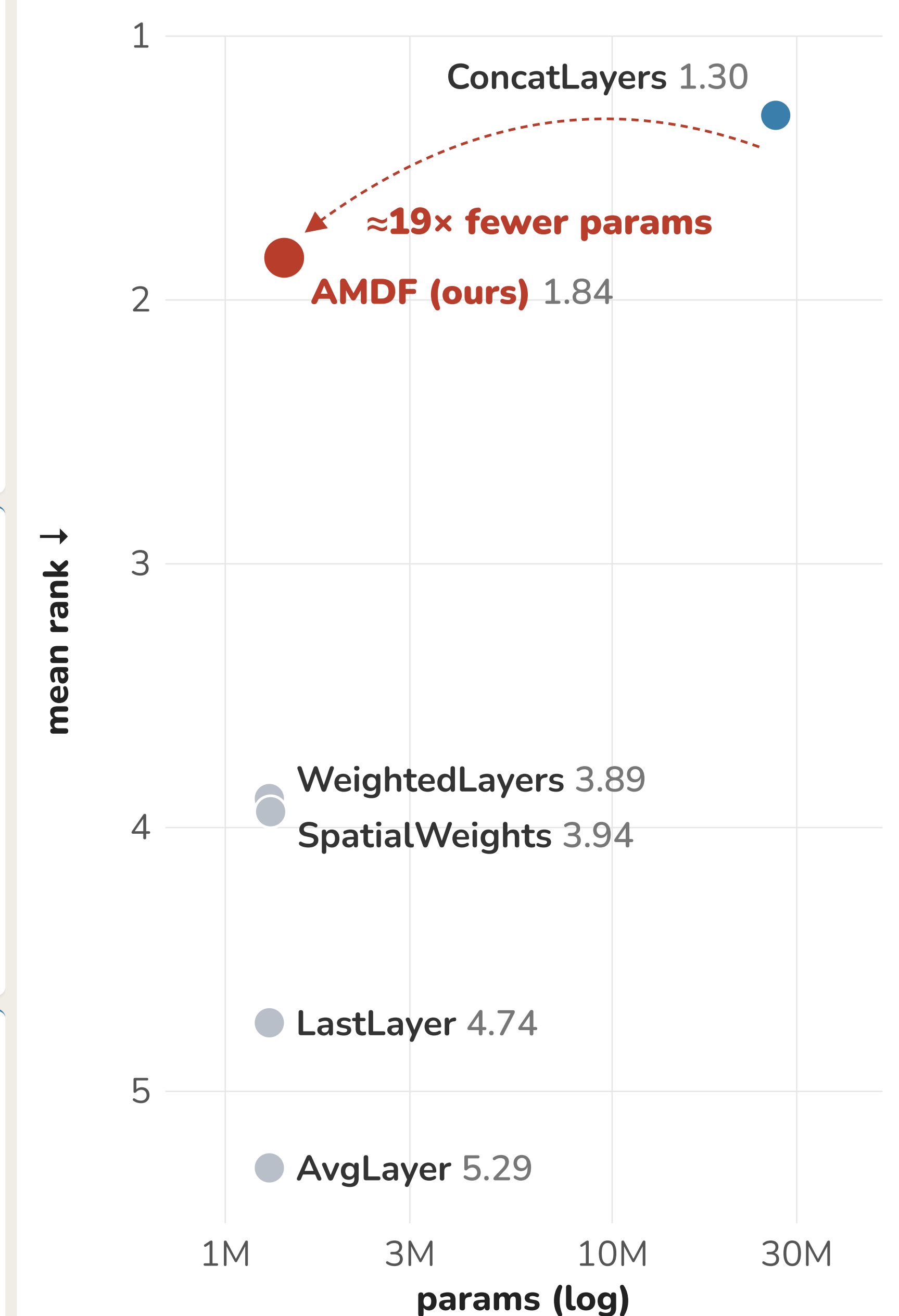


Fig. 7: ViT-L, all 5 tasks: ≈19× (pose head larger: ≈22× on dense tasks).